ASSIGNMENT-2
Linear Algebra and Calculus
Unit II: Eigenvalues and Eigenvectors

Name: _____ **Roll No.:** _____

Class: _____ **Date:** _____

Instructions:

- Attempt all questions.
- Show all necessary steps of calculation.
- Give proper justification for conceptual questions.
- Use appropriate theorems and properties wherever required.

Part A:

Q1. Characteristic Equation

Find the characteristic equation and eigenvalues of

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

Q2. Eigenvalues and Eigenvectors

Find the eigenvalues and corresponding eigenvectors of

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}.$$

Q3. Diagonal Matrix

Without expanding any determinant, write down the eigenvalues of

$$A = \begin{bmatrix} 3 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 5 \end{bmatrix}.$$

Also find an eigenvector corresponding to each eigenvalue.

Q4. Triangular Matrix and Properties of Eigenvalues

Find the eigenvalues of

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 0 & 4 & 5 \\ 0 & 0 & -1 \end{bmatrix}.$$

Hence verify that

$$\text{sum of eigenvalues} = \operatorname{tr}(A)$$

and

$$\text{product of eigenvalues} = \det(A).$$

Q5. Orthogonal Vectors

Determine whether the following pairs of vectors are orthogonal:

$$u = (1, 2, -1), \qquad v = (2, -1, 0),$$

and

$$p = (1, 1, 1), \qquad q = (1, -1, 0).$$

Give a reason for each answer.

Part B:

Q6. Eigenvalues and Eigenvectors of a 3×3 Matrix

Find all eigenvalues and corresponding eigenvectors of

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 1 \\ 0 & 1 & 3 \end{bmatrix}.$$

Q7. Properties of Eigenvalues

The eigenvalues of a 3×3 matrix A are

$$1, \qquad 2, \qquad 4.$$

Without finding the matrix explicitly, determine:

(a) the eigenvalues of A^2,
(b) the eigenvalues of A^{-1},
(c) the eigenvalues of $3A$,
(d) the eigenvalues of $A + 2I$,
(e) $\det(A)$ and $\operatorname{tr}(A)$.

Q8. Parameter-Based Question

Find the value(s) of k for which

$$A = \begin{bmatrix} 2 & k \\ 1 & 2 \end{bmatrix}$$

has equal eigenvalues.

Also find the repeated eigenvalue.

Q9. Orthogonality

Show that

$$u = (1, 1, 0), \qquad v = (1, -1, 0), \qquad w = (0, 0, 2)$$

are mutually orthogonal.

Q10. Orthogonality of Eigenvectors

Find the eigenvalues and corresponding eigenvectors of the symmetric matrix

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

Show that the eigenvectors corresponding to distinct eigenvalues are orthogonal.

Part C:

Q11. Cayley–Hamilton Theorem

State the Cayley–Hamilton theorem.

Verify the theorem for

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}.$$

Q12. Cayley–Hamilton Theorem and Inverse

Using the Cayley–Hamilton theorem, find A^{-1} for

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}.$$

Do **not** use the usual inverse formula or Gauss–Jordan method.

Q13. Higher Powers Using Cayley–Hamilton Theorem

Using the Cayley–Hamilton theorem, find A^5 for

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}.$$

Try to express A^5 in terms of A and I before calculating the final matrix.

Q14. A student claims:

> "If 0 is an eigenvalue of a square matrix A, then A must be invertible."

Is this statement true or false?

Justify your answer using the relationship between eigenvalues and the determinant of a matrix.

Also give one simple 2×2 example supporting your answer.

Q15. Consider

$$A = \begin{bmatrix} 3 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

Answer the following:

(a) Find the characteristic equation of A.

(b) Find all eigenvalues of A.

(c) Find an eigenvector corresponding to each eigenvalue.

(d) Determine whether eigenvectors corresponding to distinct eigenvalues are orthogonal.

(e) Find $\operatorname{tr}(A)$ and $\det(A)$, and verify them using the eigenvalues.

(f) Without directly multiplying A by itself, determine the eigenvalues of A^2.

Q16. Let

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

Using the Cayley–Hamilton theorem, simplify the following expression and express the answer in the form $\alpha A + \beta I$:

$$\boxed{A^8 - 5A^6 + 7A^4 - 3A^2}.$$

Q17. Let

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}.$$

Using the Cayley–Hamilton theorem, express

$$\boxed{A^{10} - 3A^8 + 5A^7 - 2A^6 + A^4}$$

in the form

$$\alpha I + \beta A + \gamma A^2.$$

Q18. Suppose that a square matrix A satisfies the equation

$$A^3 - 4A^2 + 5A - 2I = 0.$$

Using the Cayley–Hamilton theorem, reduce

$$\boxed{A^9 - 6A^8 + 12A^7 - 5A^6 + 3A^5}$$

to an expression involving only I, A, and A^2.

— End of Assignment —